Ruiwen WANG | Curriculum Vitae

ML Systems · AI Infrastructure · Distributed Computing
Paris, France | wangrw0124@gmail.com | (+33) 6 63 11 34 66
Home page | LinkedIn | Google Scholar | French citizen · Open to relocation

Production: 16,384 accelerators; TeleChat3-MoE contributor.
Open source: MindSpore / MindFormers contributions. **Research:** 7 publications · 3 patent families.

Experience

Huawei Paris Research Center Paris, France
Ph.D. Researcher (CIFRE) *Sorbonne Université & EURECOM* 2023 - 2026

- Designed symbolic cost models and hybrid-parallel planners that jointly optimise parallelism, pipeline placement and recomputation under memory constraints; benchmarked against Megatron-LM.
- Integrated planners into MindSpore / MindFormers (open-source AI training stack; SAPP-PPB), enabling production use through configuration guidance, diagnostic traces and regression benchmarks across model families and cluster scales.

Research Engineer 2022 - 2023
Research Apprentice 2021 - 2022
Distributed & Parallel Computing Lab

- Contributed to MindSpore's automatic-parallelisation framework, using symbolic cost analysis and recursive graph partitioning to generate operator-level sharding strategies for distributed DNN training.
- Profiled and tuned large-scale LLM training and inference on production Ascend 910 NPU clusters, using execution-trace analysis to diagnose compute, memory and communication bottlenecks.

Deployments & Impact

- **Scale:** Ascend 910 cluster evaluations up to 16,384 NPUs. Model coverage across projects: dense, MoE and multimodal models, including models with over 1T parameters (DeepSeek, Llama, Mistral / Mixtral, Qwen).
- **Production training:** supported China Telecom's DeepSeek-MoE deployment (Letter of Appreciation) and TeleChat3-MoE training.

Publications

First-author publications

- **Concord** - NeurIPS 2026. Model edits trigger repeated planning work. Dependency and signature checks reuse valid layer-cost evaluations across model revisions, reducing re-planning time without changing the underlying search procedure.
- **PRISM** - SCA/HPC Asia 2026. Memory feasibility depends on parallelism, scheduling and recomputation. Symbolic memory and communication models evaluate these choices without per-plan profiling, enabling pre-execution peak-memory prediction.
- **BMPipe** - IEEE CLUSTER 2025. Pipeline stages waste compute time and leave memory unused. Joint symbolic modelling and ILP optimise placement, chunking and recomputation under memory limits, improving throughput at production scale.
- **H^2O** - Euro-Par 2025, Best Poster Award. Joint planning is combinatorial. A cost-guided hierarchy couples parallelism and micro-batching with pipeline placement and recomputation, improving throughput over automated and expert-tuned plans.
- **ManuMatic** - IFIP NPC 2025. Cost models can miss runtime overlap and new kernels. A few user-specified operator shardings steer recursive partitioning and graph-consistent completion, improving throughput over the automatic baseline.

Co-authored publications

MLLM Pipeline Bubble Modeling - SCA/HPC Asia 2026 (poster).
TeleChat3-MoE - arXiv 2025 (technical report).

Expertise & Technical Skills

Parallelism: DP / TP / PP / EP; pipeline scheduling; recomputation. **Planning:** cost modelling; ILP; graph partitioning.
Performance: profiling; trace analysis; bottleneck diagnosis. **Programming:** Python, C++. **Tools:** Linux, Git.
Frameworks: PyTorch, Megatron-LM, MindSpore, MindFormers. **Systems:** Ascend CANN, HCCL.
Languages: English, French, Mandarin, Cantonese.

Patents

Execution-plan generation - WO 2025/020165 A1 (PCT), Huawei; published 2025.
Pending publication (Huawei): Large-model load balancing; Multidimensional parallelism configuration.

Education

Ph.D. Computer Science (CIFRE) - Sorbonne Université Supervisors: Raja Appuswamy, Chong Li 2026 (defended)

M.Sc. Computer Science - Sorbonne Université Advisors: Emmanuel Chailloux, Jacques Malenfant 2019 - 2022

B.Sc. Computer Science - Université Paris-Saclay Mentors: Isabelle Guyon, Viviane Pons, Zhengying Liu 2016 - 2019

Awards & Recognition

Best Poster Award - Euro-Par · Letter of Appreciation - China Telecom 2025
Golden Prize - Huawei 2024